

LLM-DR: Rule Generation with Diffusion and LLM Constraints for Temporal Knowledge Graphs

LLM-DR (Chen et al., 2025) introduces a novel method for temporal knowledge graph (TKG) extrapolation, combining diffusion-based generative models with LLM-based semantic constraints and evaluations. Designed specifically for dynamic graphs where facts evolve over time, it focuses on generating high-quality logical rules to infer future events.

Rule-Based TKG Extrapolation (With Examples)

TKGs represent facts as time-stamped quadruples:

(USA, sendsTroopsTo, Iraq, 2003)

(USA, withdrawsFrom, Iraq, 2011)

The goal is to predict future facts like:

(USA, attacks, Iraq, 2004)?

LLM-DR frames this as a reasoning task using temporal rules. A rule might look like:

Attack(X, Y, t3) $\leftarrow$ SendTroops(X, Z, t1) $\wedge$ BreakNegotiation(Z, Y, t2),
with $t3 > t1 \geq t2$

These rules are first mined from data using temporal path sampling (e.g., walking from USA $\rightarrow$ Iraq with specific relations at specific times), then used as seeds for generation.

Diffusion-Based Rule Generation (With Examples)

Instead of directly using existing rules, LLM-DR adds noise to the rule embeddings to form a **forward diffusion process**, then denoises them in reverse to generate new rules.

Example:

Start from a rule:

(Consult(X, Y, t3) $\leftarrow$ SendEnvoy(X, Z, t1) $\wedge$ ExpressIntent(Z, Y, t2))

$\rightarrow$ Add noise $\rightarrow$ recover new rule:

(Negotiate(X, Y, t3) $\leftarrow$ MakeStatement(X, Z, t1) $\wedge$ Visit(Z, Y, t2))

This allows generation of rules that didn't explicitly appear in the data but follow statistical and temporal patterns.

LLM-Based Constraints for Semantic Filtering (With Examples)

To prevent irrelevant or nonsensical rules, the model queries an LLM to restrict which relations are allowed in a rule body.

Example Prompt:

"Given the rule head **Consult**, select k most semantically related relations."

LLM Output:

- Engage in negotiation
- Request mediation
- Express willingness to cooperate

These relations are then allowed in the rule body for that head. Rules like
 $\text{Consult}(X,Y,t3) \leftarrow \text{WatchMovie}(X,Z,t1)$
 are rejected.

Coarse-to-Fine Rule Evaluation (With Examples)

Once rules are generated, they are evaluated in two stages:

1. Coarse-grained via LLM (Chain-of-Thought)

The LLM explains whether the rule makes sense logically and temporally.

Example:

Rule:

$\text{Criticize}(X,Y,t3) \leftarrow \text{Investigate}(X,Z,t1) \wedge \text{AppealForAid}(Z,Y,t2)$

LLM: “Implausible. There is no direct connection between aid and criticism.”

2. Fine-grained via BERT (Semantic Scoring)

Embed rule head and rule body and compute cosine similarity.

Example:

Head: Accuse

Body: ExpressDoubt $\wedge$ Blame

→ Cosine similarity score: 0.85 (high)

Implausible rules may have low similarity (e.g., 0.35)

Final Use in Prediction (With Example)

After filtering, these logical rules are used to infer missing facts.

Given:

$(\text{USA}, ?, \text{Iraq}, 2004)$

The model might apply the rule:

$\text{Attack}(X,Y,t3) \leftarrow \text{SendTroops}(X,Z,t1) \wedge \text{BreakNegotiation}(Z,Y,t2)$

→ Finds $\text{SendTroops}(\text{USA}, \text{Kuwait}, 2003)$ and $\text{BreakNegotiation}(\text{Kuwait}, \text{Iraq}, 2003)$

→ Predicts $\text{Attack}(\text{USA}, \text{Iraq}, 2004)$

Evaluation and Results

Tested on ICEWS and GDELT datasets for TKG link prediction, LLM-DR outperforms:

- Embedding models (RE-GCN, CyGNet)
- Rule-based models (TLogic, TR-Rules)
- LLM-only methods (Vicuna, LLaMA)

Ablation shows:

- Removing diffusion rule generation → large performance drop
- Replacing BERT with LLM for scoring → worse semantic accuracy
- No constraints → more junk rules

Why It Matters for the Survey

LLM-DR is one of the first methods to **blend generative modeling (diffusion)** with **LLM-guided symbolic reasoning** over temporal graphs. Its hybrid pipeline allows controllable, interpretable rule generation with semantic quality filters. This positions LLM-DR as a key contribution in the emerging intersection of **LLMs, dynamic graph modeling, and logical reasoning**.